Experiment No. 9

Title

Client-Side Storage using localStorage and sessionStorage

Aim

To implement localStorage and sessionStorage in JavaScript to save and manage user preferences such as theme settings.

Software / Tools Required

- Visual Studio Code (VS Code)
- Google Chrome
- HTML
- CSS
- JavaScript

Theory

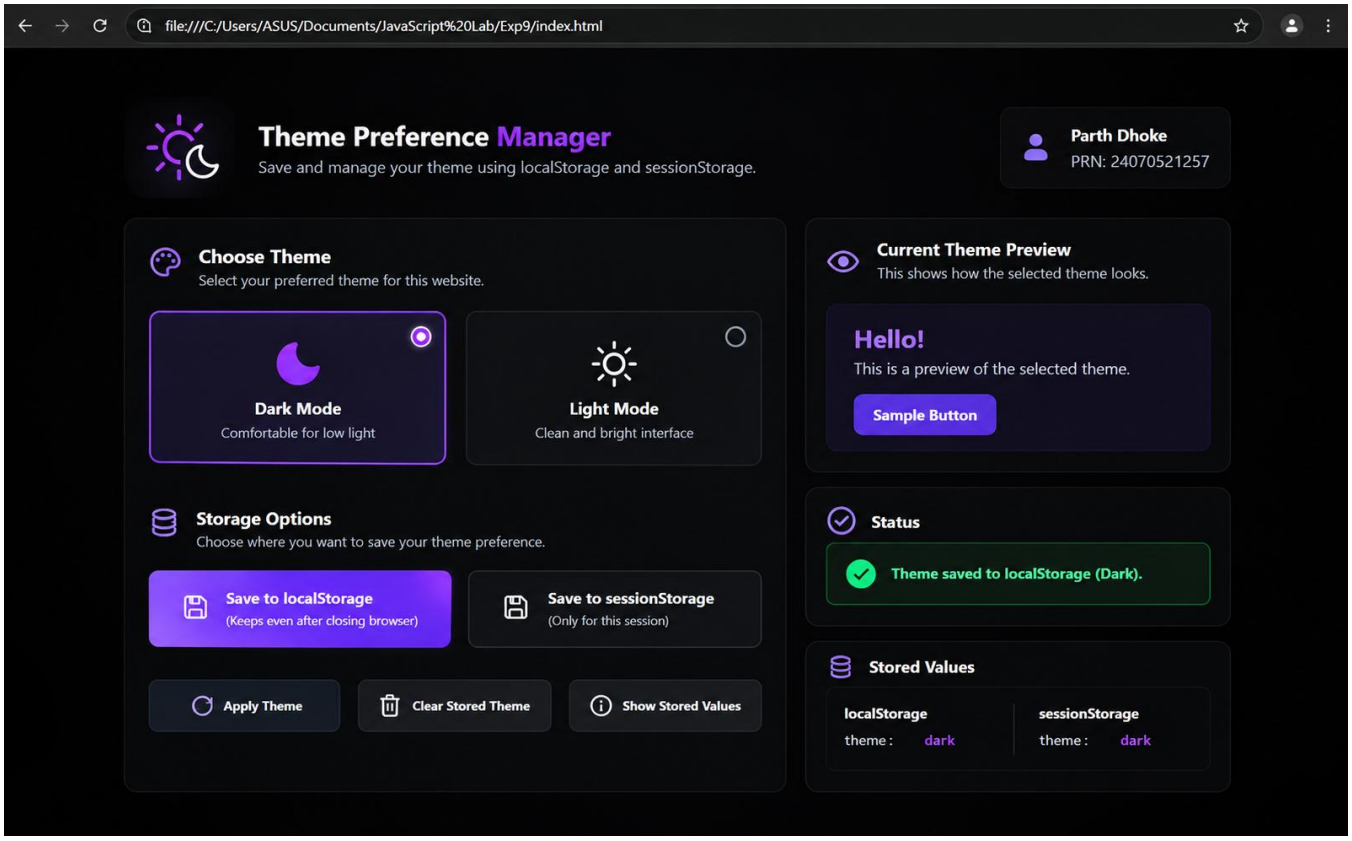
This experiment explores HTML5 Web Storage for storing and managing user preferences on the client side. The application demonstrates the use of localStorage and sessionStorage to save the user's selected theme and restore the preference when the webpage is loaded.

- **Web Storage:** HTML5 provides Web Storage APIs that allow web applications to store key-value data in the user's browser without requiring a server-side database.
- **localStorage:** The application uses localStorage to permanently store the selected theme preference. The stored preference remains available even after closing and reopening the browser.
- **sessionStorage:** The application also uses sessionStorage to store the selected theme for the current browser session. The stored data is available while the session remains active.
- **Storing Data:** Methods such as localStorage.setItem() and sessionStorage.setItem() are used to save the user's selected theme preference.
- **Retrieving Data:** The methods localStorage.getItem() and sessionStorage.getItem() are used to retrieve previously stored theme preferences and restore the corresponding interface.
- **Removing Stored Data:** The removeItem() and clear() methods can be used to remove stored preferences from Web Storage.
- **Dynamic Theme Switching:** JavaScript dynamically changes the webpage between Light and Dark themes according to the user's selection.

Output

Theme Settings Web Application

The developed webpage provides a theme selection interface where the user can select a preferred Light or Dark theme and save the preference using either localStorage or sessionStorage.



Result / Conclusion

In conclusion, this experiment demonstrated the implementation of HTML5 Web Storage using localStorage and sessionStorage through a functional Theme Settings application. The application successfully stored the user's theme preference, retrieved previously saved preferences, dynamically switched between light and dark themes, and provided appropriate feedback to the user. Thus, the experiment demonstrated effective use of client-side storage and JavaScript for maintaining user preferences in a web application.